

Your grade: 80%

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1. A Transformer Network, unlike its predecessors RNNs, GRUs and LSTMs, can process entire sentences all at the same time. (Parallel architecture).

1 / 1 point

☒ True

☐ False

✓ **Correct**

A Transformer Network can ingest entire sentences all at the same time.

2. Transformer Network methodology is taken from: (Check all that apply)

1 point

☐ None of these.

☐ Convolutional Neural Network style of architecture.

☐ Convolutional Neural Network style of processing.

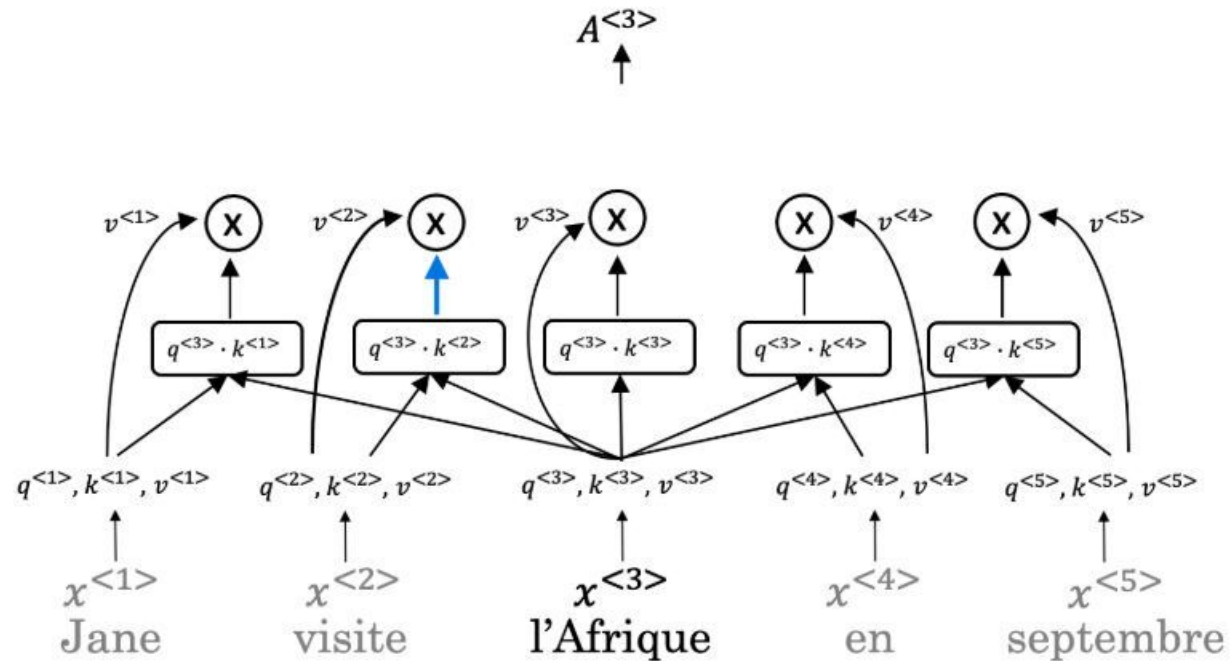
☒ Attention mechanism.

✓ **Correct**

You didn't select all the correct answers

3. How does the Self-Attention mechanism of transformers use neighboring words to compute a word's context?

1 point



- ☒ Multiplication of the word values to map the Attention related to that given word.
- ☐ Selecting the maximum word values to map the Attention related to that given word.
- ☐ Selecting the minimum word values to map the Attention related to that given word.
- ☐ Summation of the word values to map the Attention related to that given word.

- ☒ Multiplication of the word values to map the Attention related to that given word.
- ☐ Selecting the maximum word values to map the Attention related to that given word.
- ☐ Selecting the minimum word values to map the Attention related to that given word.
- ☐ Summation of the word values to map the Attention related to that given word.

✗ **Incorrect**

Multiplication is not used to compute a word's context. To revise the concept watch the lecture *Self-Attention*.

4. Which of the following correctly represents *Attention* ?

1 / 1 point

- ☒ $Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$
- ☐ $Attention(Q, K, V) = softmax(\frac{QV^T}{\sqrt{d_k}})K$
- ☐ $Attention(Q, K, V) = min(\frac{QV^T}{\sqrt{d_k}})K$
- ☐ $Attention(Q, K, V) = min(\frac{QK^T}{\sqrt{d_k}})V$

✓ **Correct**

1 / 1 point

5. Are the following statements true regarding Query (Q), Key (K) and Value (V) ?

Q = interesting questions about the words in a sentence

K = qualities of words given a Q

V = specific representations of words given a Q

☒ True

☐ False

✓ **Correct**

Q = interesting questions about the words in a sentence, K = qualities of words given a Q, V = specific representations of words given a Q

$$\text{Attention}(W_i^Q Q, W_i^K K, W_i^V V)$$

1 / 1 point

6. i here represents the computed attention weight matrix associated with the i th “head” (sequence).

☐ False

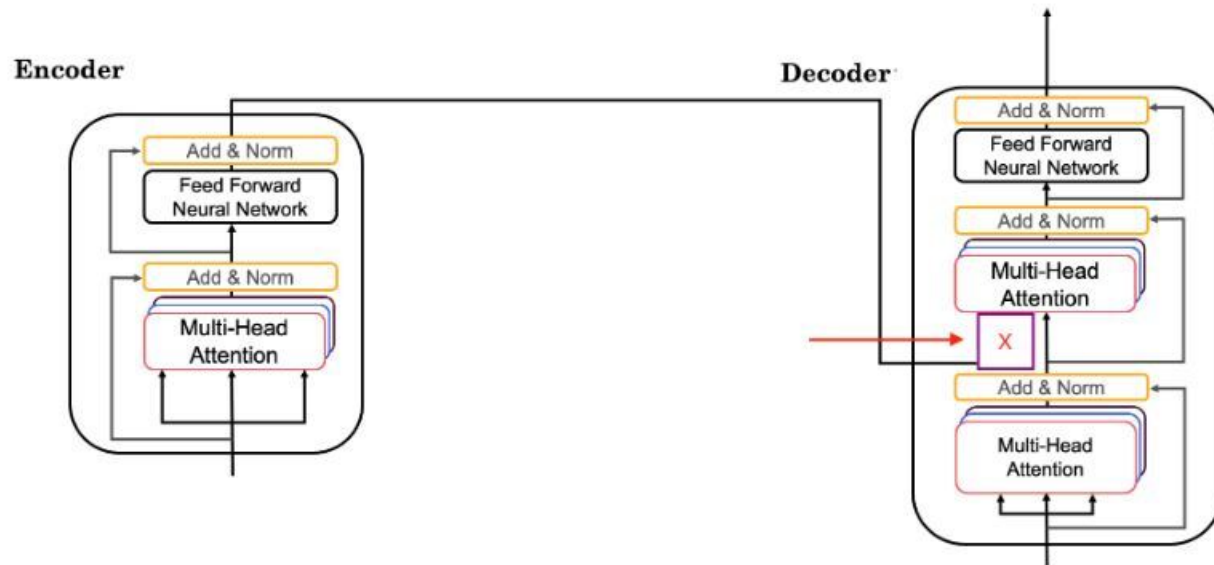
☒ True

✓ **Correct**

i here represents the computed attention weight matrix associated with the i th “head” (sequence).

7. Following is the architecture within a Transformer Network (*without displaying positional encoding and output layers(s)*).

1 / 1 point



What information does the *Decoder* take from the *Encoder* for its second block of *Multi-Head Attention* ?
(Marked X , pointed by the independent arrow)

(Check all that apply)

☐ Q

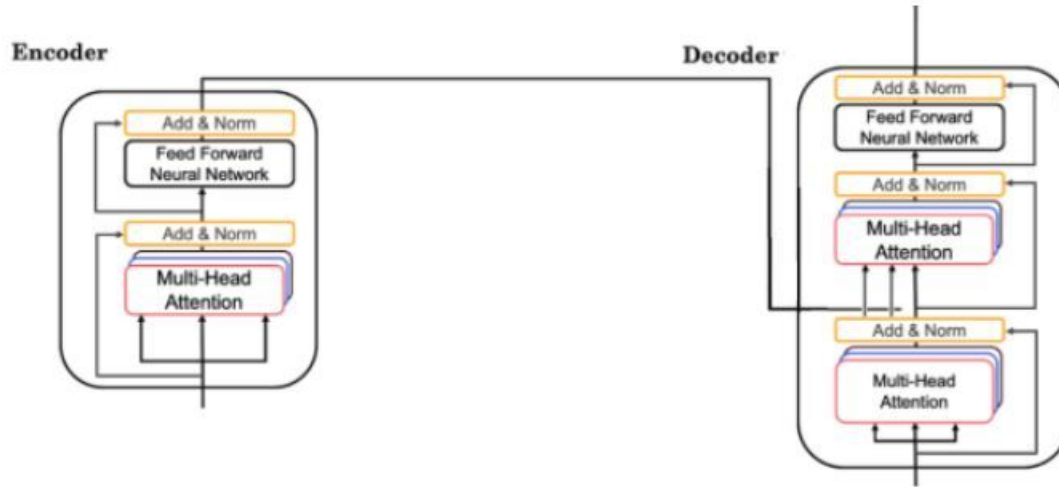
☒ K

✔ Correct

☒ V

✔ Correct

8. Following is the architecture within a Transformer Network (*without displaying positional encoding and output layers(s)*).



The output of the decoder block contains a softmax layer followed by a linear layer to predict the next word one word at a time.

- ☐ True
- ☒ False

✓ **Correct**

The output of the decoder block contains a linear layer followed by a softmax layer to predict the next word one word at a time.

9. Which of the following statements is true?

1 / 1 point

- ☐ The transformer network is similar to the attention model in that both contain positional encoding.
- ☒ The transformer network differs from the attention model in that only the transformer network contains positional encoding.
- ☐ The transformer network differs from the attention model in that only the attention model contains positional encoding.
- ☐ The transformer network is similar to the attention model in that neither contain positional encoding.

✓ **Correct**

Positional encoding allows the transformer network to offer an additional benefit over the attention model.

10. Which of these is a good criterion for a good positional encoding algorithm?

1 / 1 point

☒ It should output a unique encoding for each time-step (word's position in a sentence).

☒ Correct

☒ Distance between any two time-steps should be consistent for all sentence lengths.

☒ Correct

☒ The algorithm should be able to generalize to longer sentences.

☒ Correct

☐ None of these.